Department of Nano Science Technology
Faculty of Technology
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Centralized Workshops & Lab Scheduling Platform

SQL scripts:CREATE TABLE statements and sample INSERT data

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-- SQLite Foreign Key constraints activate
PRAGMA foreign_keys = ON;

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-- 1. CREATE TABLE STATEMENTS (Aligned with ER Diagram)
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-- Table: students
CREATE TABLE IF NOT EXISTS students (
    index_no VARCHAR(20) PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    department VARCHAR(50) NOT NULL,
    email VARCHAR(100) NOT NULL UNIQUE
);

-- Table: workshops
CREATE TABLE IF NOT EXISTS workshops (
   category VARCHAR(50) PRIMARY KEY,
    department VARCHAR(50) NOT NULL
);

-- Table: records
CREATE TABLE IF NOT EXISTS records (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    index_no VARCHAR(20) NOT NULL,
    category VARCHAR(50) NOT NULL,
    supervisor_name VARCHAR(100) NOT NULL,
    project_name VARCHAR(100) NOT NULL,
    record_date DATE NOT NULL,
    description TEXT,
    passcode VARCHAR(50) NOT NULL UNIQUE,
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    created_at TIMESTAMP NOT NULL,
    status VARCHAR(20) NOT NULL DEFAULT 'Pending' CHECK(status IN ('Pending', 'Approve', 'Reject')),
    admin_remarks TEXT,
    FOREIGN KEY (index_no) REFERENCES students(index_no),
    FOREIGN KEY (category) REFERENCES workshops(category)
);

-- Table: booking_slots
CREATE TABLE IF NOT EXISTS booking_slots (
    record_id INT NOT NULL,
    time_slot VARCHAR(50) NOT NULL CHECK(length(trim(time_slot)) > 0 AND instr(time_slot, ',') = 0),
    PRIMARY KEY (record_id, time_slot),
    FOREIGN KEY (record_id) REFERENCES records(id) ON DELETE CASCADE
);

-- Index for optimizing date and category search queries
CREATE INDEX IF NOT EXISTS idx_records_category_date ON records(category, record_date);



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-- 2. SAMPLE INSERT DATA
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-- Inserting into students
INSERT INTO students (index_no, name, department, email) VALUES
('ET/2021/045', 'Keshan Abesinghe', 'Material & Nano Science Technology',
'keshan.abesinghe@gmail.com'),
('ET/2021/012', 'Nimal Jayasinghe', 'Mechanical Engineering Technology', 'nimal.j@gmail.com'),
('ET/2021/089', 'Kavindi Perera', 'Electrical & Electronic Technology', 'kavindi.p@gmail.com');

-- Inserting into workshops
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INSERT INTO workshops (category, department) VALUES

('Carpentry & Woodwork Lab', 'Mechanical Engineering Technology'),

('CNC & Machining Center', 'Mechanical Engineering Technology'),

('Advanced Materials & Nano Lab', 'Material and Nano Science Technology'),

('Electronics & Embedded Systems Lab', 'Electrical & Electronic Technology');


-- Inserting into records

INSERT INTO records (id, index_no, category, supervisor_name, project_name, record_date, description, passcode, created_at, status, admin_remarks) VALUES

(1, 'ET/2021/045', 'Advanced Materials & Nano Lab', 'Dr. S. Wijesinghe', 'Graphene Synthesis & Analysis', '2026-10-20', 'Spectrophotometer and sample extraction analysis', 'PASS-1001', '2026-10-13 09:30:00', 'Approve', 'Approved with lab safety compliance'),

(2, 'ET/2021/012', 'Carpentry & Woodwork Lab', 'Eng. M. Bandara', 'Autonomous Rover Wooden Chassis', '2026-10-21', 'Chassis base cutting and milling work', 'PASS-1002', '2026-10-13 10:15:00', 'Pending', NULL),

(3, 'ET/2021/089', 'Electronics & Embedded Systems Lab', 'Dr. P. Fernando', 'Smart Energy Monitor Circuit', '2026-10-20', 'Oscilloscope testing and PCB assembly', 'PASS-1003', '2026-10-13 11:00:00', 'Reject', 'Scheduled maintenance during this slot');


-- Inserting into booking_slots

-- Record 1 (Multi-slot booking)

INSERT INTO booking_slots (record_id, time_slot) VALUES

(1, '8.30 am - 10.30 am'),

(1, '10.30 am - 12.30 pm');


-- Record 2 (Single slot booking)

INSERT INTO booking_slots (record_id, time_slot) VALUES

(2, '1.30 pm - 3.30 pm');


-- Record 3 (Single slot booking)

INSERT INTO booking_slots (record_id, time_slot) VALUES

(3, '3.30 pm - 5.30 pm');
```